

Development of the Module of Charging Stations Placement for Electric Transport Based on Genetic Algorithm

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Abstract—The implementation of energy transformation strategies has led to tangible changes in the approaches to the organization of modern energy systems through the involvement of new generation sources and new types of consumers that require better planning and organization of distributed systems. One example of such innovations is the ubiquity of electric transportation in large cities, which is becoming widespread. This process is driven by various factors, the most important of which is the environmental safety factor of modern society. Despite the fact that the development of technology makes such transport more and more capable and competitive with traditional types, there are some limitations, primarily due to the low autonomy of such vehicles, which leads to the need to organize specialized charging points. The problem of charging station location involves determining the most optimal locations for station placement, depending on the selected parameters of the target function, for which various approaches are used: linear programming, multi-criteria and meta-heuristic optimization, genetic and evolutionary algorithms. This paper considers the process of implementation of the software module for determining the optimal locations of electric vehicle charging stations based on the use of genetic algorithm, the main criterion of which is the maximum utilization rate of the operated stations.

Keywords—genetic algorithms, optimization, electric transport, charging stations

I. INTRODUCTION

In recent years, electric vehicles have become widespread (Fig. 1), which is facilitated by the development of energy [1] storage technologies and the emergence of new designs of electric motors that make it possible to ensure high competitiveness of this type of vehicles [2].

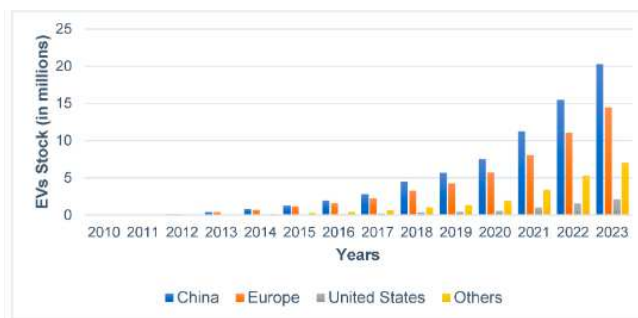


Fig. 1. Global electric car stock country-wise, including both battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs) [3]

Several major factors contribute to the active adoption of personal and public electric transportation:

- Environmental friendliness. Electric cars do not produce greenhouse gases, which are one of the main causes of harmful effects on the climate [4], [5];
- High energy efficiency. Electric vehicles have higher efficiency than traditional internal combustion engines, allowing for more efficient use of energy and lower overall “fuel” costs and lower overall operating costs [6];
- Reduced dependence on hydrocarbon resources. A greater share of electric vehicles in total transportation reduces economic and geopolitical risks and increases energy security;

Reduction of noise and vibrations. The use of electric vehicles allows to somewhat smooth out the general “sound background” [7], which is an important factor influencing human health.

At the same time, with the introduction and growth of electric vehicles, a number of challenges arise related to the creation and optimization of an appropriate energy infrastructure [8]:

- organization of intelligent control systems for distributed energy infrastructure;
- ensuring the sustainability of electricity supply;
- organization of effective integration into the existing energy system.

These tasks can be attributed to the second phase of efficient utilization of electric vehicles, while the first priority is to identify the proposed locations of charging stations to cover the existing demand.

In this connection, the paper considers the process of implementation of the module of electric vehicle charging station placement based on the use of genetic algorithm, which takes into account two main factors of optimization of the final solution: the parameter of utilization of operating stations, and the use of priority points (parking zones) used as an initial set of potential coordinates for station placement.

II. MATERIALS AND METHODS

The problem of determining the proposed location of the charging station is a well-known problem, for the solution of which various approaches and methods of optimization are used, such as: genetic and evolutionary algorithms, neural networks [9], multi-criteria optimization methods [10], the classification of which is presented in Fig. 2.

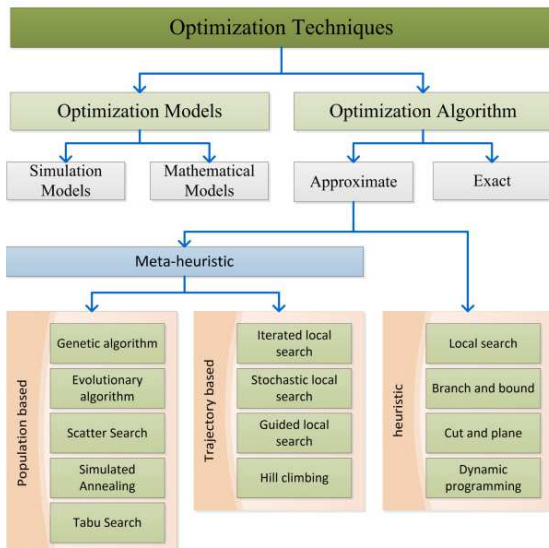


Fig. 2. Classification of optimization techniques

After selecting the proposed optimization method, there is a task of forming input parameters, which are some spatial data, defining the constraint system and the key optimization parameter. There are two main approaches to solving such problems:

- object-location configuration implies that optimization is performed on a certain set of predefined locations [11];
- A “site-to-site” formulation, implying that charging stations are freely available within a certain application area to cover existing needs [12].

The second approach is more complex, requiring more input parameters that cannot always be obtained from public data sources, so we will use the first formulation of the object-location optimization problem.

Thus, it is necessary to define a preliminary set of assumed location coordinates, information about which can be obtained by using the OpenStreetMap tool [13]**Error! Reference source not found..** The obtained set may be excessive, therefore, for efficient operation of the optimization algorithm, we obtain a part of the initial values using the values of the charging station coverage radius and the Haversine formula [14][13]. The general algorithm of the whole program is shown in Fig. 3.

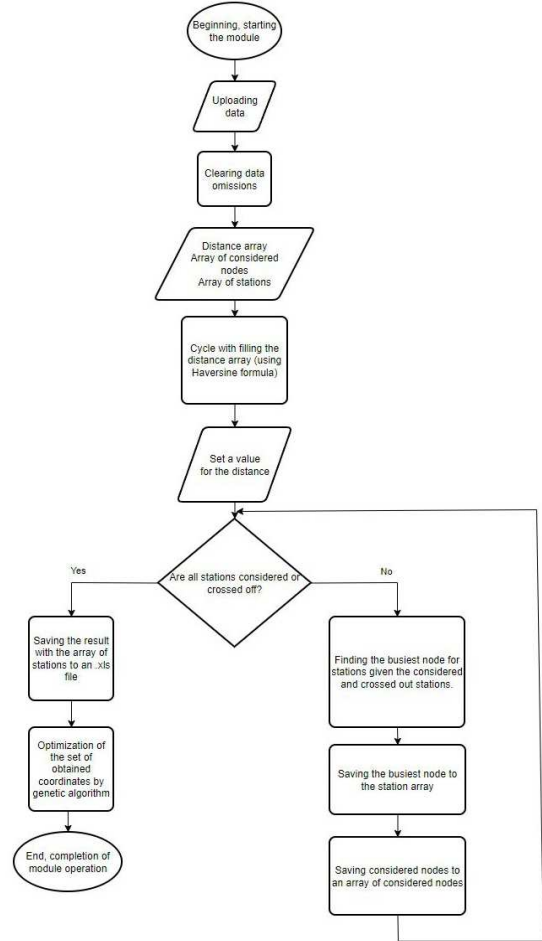


Fig. 3. General algorithm of the program

III. PRACTICAL SIGNIFICANCE

Since utilization rate is considered as the main efficiency criterion for the placement of charging stations, it is necessary to select a set of key locations to reach as many potential consumers as possible. One such site could be parking areas, which are likely to have additional space to accommodate charging infrastructure. The area of each facility was used as a weighting factor, resulting in areas with a larger area being prioritized.

The city of Krasnodar was chosen as an example for the study, the map with the location of parking lots of which is shown in Figure 4a. Such a large number of parking lots can aggravate the use of the genetic algorithm, making its running time extremely long and the algorithm itself extremely demanding on resources [15]. To solve this problem, the most relevant parking zones with the largest area and with a predetermined radius of location were

selected, for which a distance matrix was constructed by using the Haversine formula (Fig. 4b).

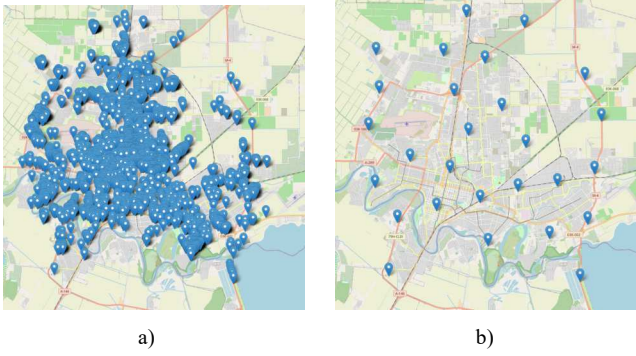


Fig. 4. Visualization of parking spaces: a) all available parking areas; b) selection of the most suitable spaces (by area value)

The obtained coordinates are the input parameters for the used genetic algorithm, the algorithm of which is presented in Fig. 5.

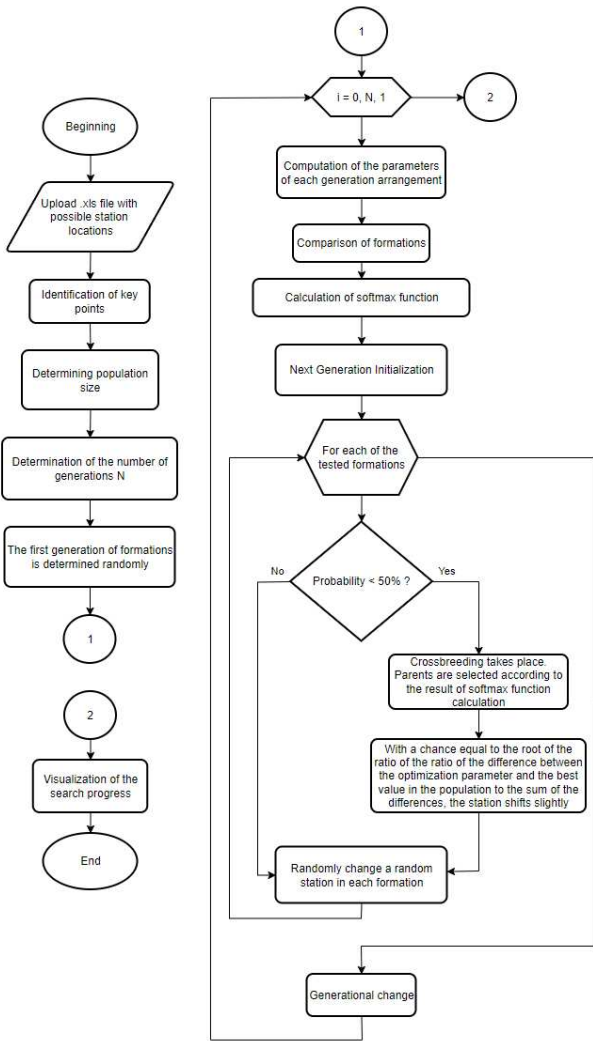


Fig. 5. Block diagram of the genetic algorithm for determining optimal station locations

After the first generation of formations is determined, the main cycle of the genetic algorithm is started, within which the parameters for each formation are calculated, the current best one is found, and all available formations are compared with it, which is necessary for realization of the procedure of softmax function calculation.

The use of this algorithm allows to optimize the initial obtained station placement, taking into account the criterion of station utilization. Fig. 6 shows the result of using the optimization algorithm on the input set of proposed locations.

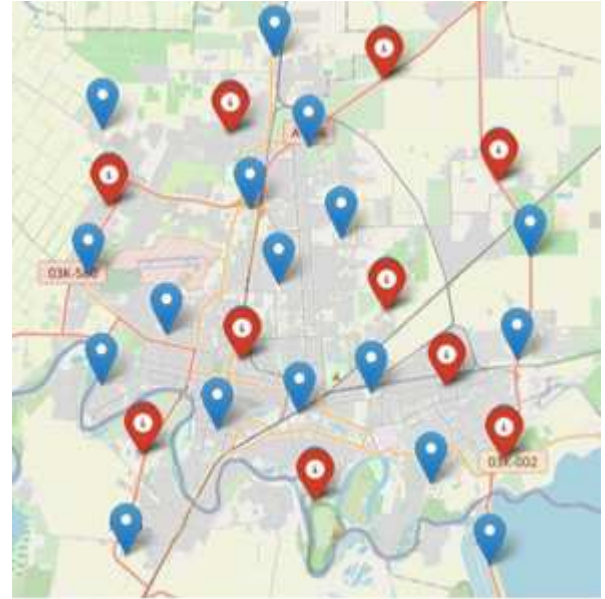


Fig. 6. The result of the genetic algorithm

IV. CONCLUSION

The implemented station placement module can be used to determine the optimal placement of charging stations based on a set of starting points and a known estimated number of stations to be placed. The proposed approach is somewhat idealized as it does not fully take into account all parameters and factors that influence the success of a charging station. As a next step in development, additional data sets, such as road traffic and its importance, the number of residential buildings in the district, should be involved to allow a more realistic assessment of the available charging infrastructure needs.

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